B

Balancing; Onspeed Balancing of a Rotor

Balancing generally refers to the balancing of a turbomachinery rotor. Balancing can, in some cases, be done in the field. Maintenance staff can be trained, for instance, to balance a pump in situ in the plant, if their readings with their vibration analysis equipment confirm that this is what needs to be done. For more critical items, such as process compressors, this process is best done in the overhaul facility of the original equipment manufacturer (OEM). The exception to this would be if the end user had his or her own balance equipment and had trained staff that was capable of handling the rotor in question.

Most balancing of rotating machinery rotors or components of rotors (such as turbine wheels and so forth) is done in a balancing machine at speeds in the neighborhood of 1800 to 2000 r/min in atmospheric conditions. In certain rare instances, balancing at these speeds does not remove the imbalance (that was causing rotor vibration in the first place). As a last resort, the process engineer may have to specify onspeed balancing of this rotor. This needs to be done in a vacuum chamber and is expensive. Also, there are very few suitable vacuum test facilities in the world. Before getting into this additional expense, the process engineer is probably well advised to consult a rotating machinery engineer.

Balancing Problems, Troubleshooting (Turbomachinery) (*see* Condition Monitoring)

Batteries (*see* Cells)

Bearings* (*see also* Lubrication)

Bearings permit relative motion to occur between two machine elements. Two types of relative motion are possible, rolling or sliding, each of which depends upon the design of the mechanical bearing element. Thus bearings are classified into two general types: the rolling-contact type (rolling) and the sliding-contact bearing design in which the bearing elements are separated by a film of oil (sliding). Both can be designed to accommodate axial and/or radial loads. Each has a wide variety of types and designs to fit a wide variation in uses. The selection of a bearing type for application to a particular situation involves a performance evaluation and cost consideration.

There is ample literature available to determine the relative merits of each type. This subsection will provide a general overview of the types of bearings presently encountered in the equipment covered in this edition. Changes take place frequently because many scientists and engineers work constantly to improve the state of the art in bearing design.

* Source: Demag Delaval, USA.

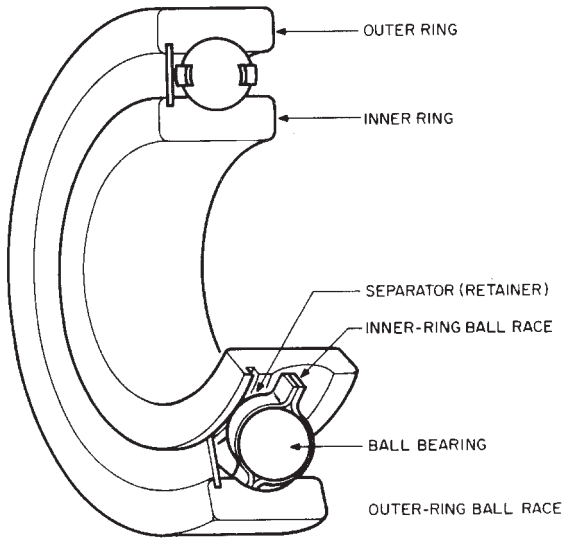


FIG. B-1 Nomenclature of a ball bearing. (Source: Demag Delaval.)

Rolling-Contact Bearings

Rolling-contact bearings include ball bearings, roller bearings, and needle bearings. Within each category several variations have been developed for specific applications. Variations in the amounts of radial and thrust load capabilities also exist between specific types. Self-aligning ball or roller bearings, by virtue of their spherically ground outer race, can tolerate misalignment of the shaft or housing.

Rolling-contact bearings consist of four principal components: an outer race, an inner race, rolling elements, and a separator, or spacer, for the rolling elements. The inner ring is mounted on the shaft. The outer ring securely fits in a stationary housing. The facing surfaces of the inner and outer rings are grooved to conform to the rolling-element shape. The rolling elements (with separator) accurately space the inner and outer races and thus enable smooth relative motion to occur (see Fig. B-1).

Sliding-Contact Bearings

Sliding-contact bearings are classified into two general types: journal bearings and thrust bearings. Journal bearings support radial loads imparted by the rotating shaft and may also be required to arrest or eliminate hydraulic instabilities that may be encountered in lightly loaded high-speed machinery. The thrust bearings are used for loads parallel to the shaft and may be required to support the full weight of the rotor in cases of vertical machinery.

Journal Bearings

The common types of journal bearings are:

- Plain journal bearings
- Three-lobe journal bearings
- Tilting-pad journal bearings

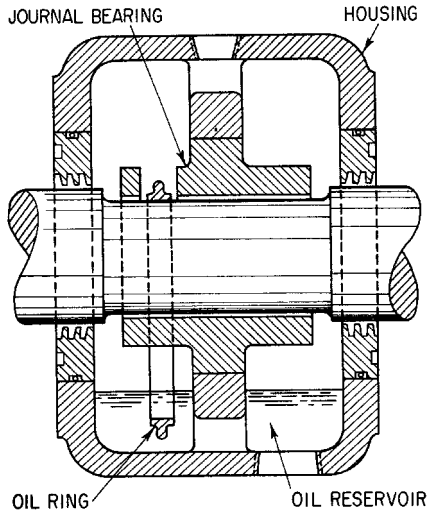


FIG. B-2 Ring-oil lubrication. (Source: Demag Delaval.)

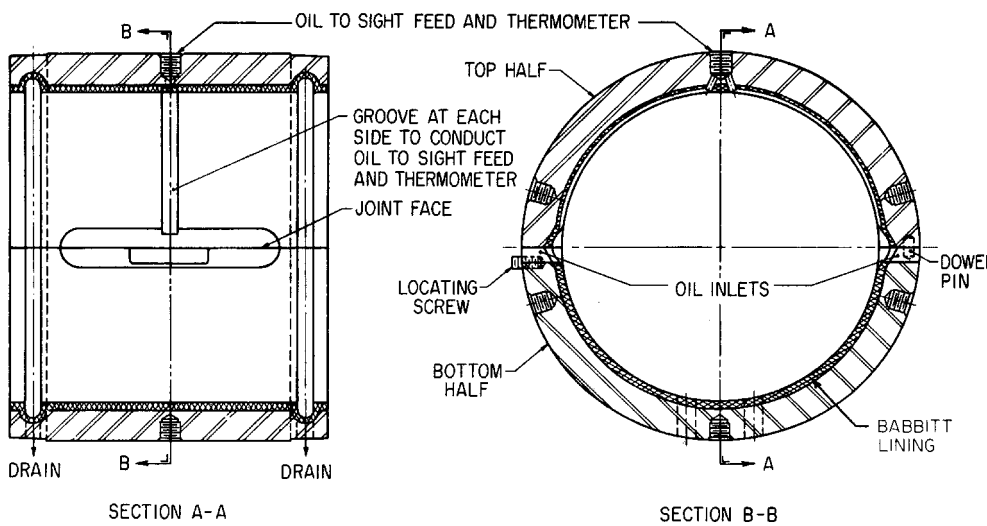


FIG. B-3 Typical design of journal bearing. (Source: Demag Delaval.)

The *plain journal bearing* may be either force-fed from a pressure-lubrication system or ring-oiled by means of a free ring that rests on and rotates with the shaft to serve as a pumping medium, as shown in Fig. B-2. A section of a typical force-fed journal bearing is shown in Fig. B-3. This shows a babbitt-lined split-type bearing. Journal-bearing clearance for high-speed machinery (i.e., turbines, centrifugal compressors, pumps, etc.) should be not less than 0.0015 in/in of diameter for satisfactory operation. However, a complete analysis should be made to obtain the proper clearance for each case.

The *three-lobe journal bearing* shown in Fig. B-4 is applied on high-speed machinery, usually when the bearing is end-fed and used in combination with a face-contact seal or when the bearing itself is a shaft seal. The three-lobe design provides, intrinsically, a stable three-point support that is particularly adaptable to very light loads in either horizontal or vertical shafts. Because of the small

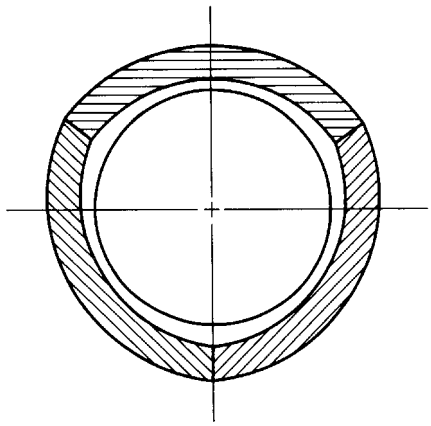


FIG. B-4 Sketch of a three-lobe journal bearing. (Source: Demag Delaval.)

amount of additional clearance, the end-leakage effect of this bearing can be well controlled and provides good impurity tolerance from the oil system with a means for continuous flushing.

Another design of a three-lobe bearing has three equal-diameter bores that are displaced 120° from each other by an amount equal to the bearing clearance less the oil-film thickness. This bearing is center-fed. It is used to support lightly loaded shafts, especially light gear shafts in which the load direction varies with the transmitted torque.

The *tilting-pad journal bearing* consists of three or more separate shoes equally spaced around the circumference of the journal. Its mechanical design and performance follow closely the theory of the tilting-pad thrust bearing. A typical bearing of this kind utilizes five pads, of which one pad is on the bottom, dead center for horizontal shafts, to provide a fixed means of supporting the shaft during alignment. This bearing is considered to be the best design for attaining rotor stability. It also is inherently more tolerant of impurities in the oil system without bearing failure. A typical section of this bearing is shown in Fig. B-5. This bearing is pressure-fed from a circulation system, and the amount of oil passing through the bearing is controlled by a restriction at the inlet to the bearing shell.

Thrust Bearings

The simplest type of thrust bearing is the sliding-contact type with plain babbitted face, which is used in lightly loaded thrust applications. The second, more complicated design for intermediate thrust loading is found in the tapered-land thrust bearing. This design has a multiple radial series of fixed lands having a tapered inlet edge in the direction of rotation to develop a hydrodynamic film to support the thrust load.

The most common thrust bearing found in high-speed machinery is the tilting-pad thrust bearing. This bearing contains independent-acting shoes with a hardened, pivoted pad backing. Each shoe acts to establish a hydrodynamic film proportional to the speed and loading of the shaft to support the thrust load. This bearing may be single-sided in the case of turbines in which unidirectional thrust is encountered or double-sided in the case of compressors or pumps in which the thrust loading may shift, depending on the operating conditions.

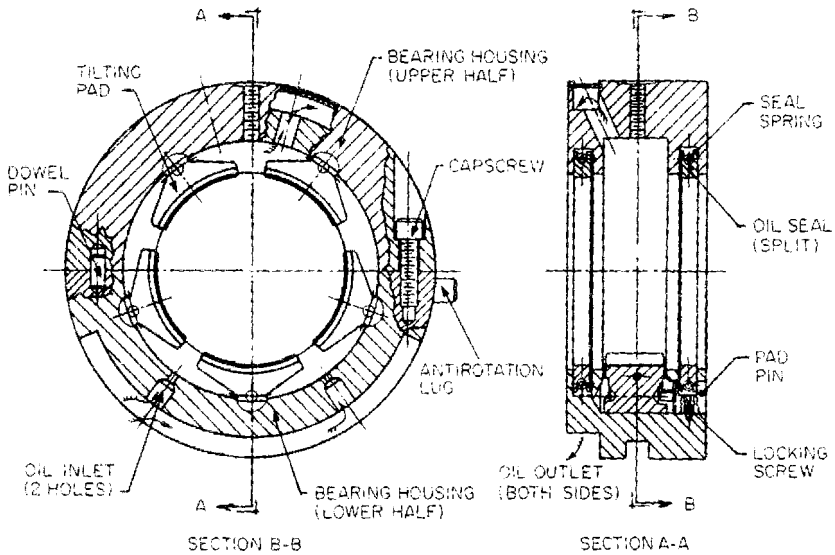


FIG. B-5 Typical tilting-pad bearing. (Source: Demag Delaval.)

Magnetic Bearings*

Magnetic bearing systems represent a completely different approach to the support of rotating equipment. They are a *noncontacting technology*, which means negligible friction loss and wear and higher reliability. It also enables previously unachievable surface speeds to be attained. Lubrication is eliminated, meaning that these bearings can be incorporated into processes that are sensitive to contamination, such as the vacuum chambers in which many semiconductor manufacturing processes take place.

Magnetic bearings incorporate three distinct technologies: *actuators and sensors* are the electromechanical hardware by which input signals are collected and supporting forces applied to the machine on which they are installed. The *control system* provides the power and control electronics for signal conditioning prior to calculation of correcting forces and resultant commands to the power amplifiers for each axis of control. *Control algorithms* are the software programs used in digital magnetic bearing system control including the processing of the input signals after conditioning and calculation of the command signals to the power amplifiers.

System components and operation

Magnetic bearings provide *attractive electromagnetic suspension* by application of electric current to ferromagnetic materials used in both the stationary and rotating parts (the *stator* and *rotor*, respectively) of the magnetic bearing *actuator*. This creates a flux path that includes both parts, and the air gap separating them, through which noncontact operation is made possible.

As the air gap between these two parts decreases, the attractive forces increase; electromagnetic bearings are inherently unstable. A *control system* (Fig. B-6) is needed to regulate the current and provide stability of the forces and rotor position.

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*Source: Revolve, Canada. Adapted with permission.

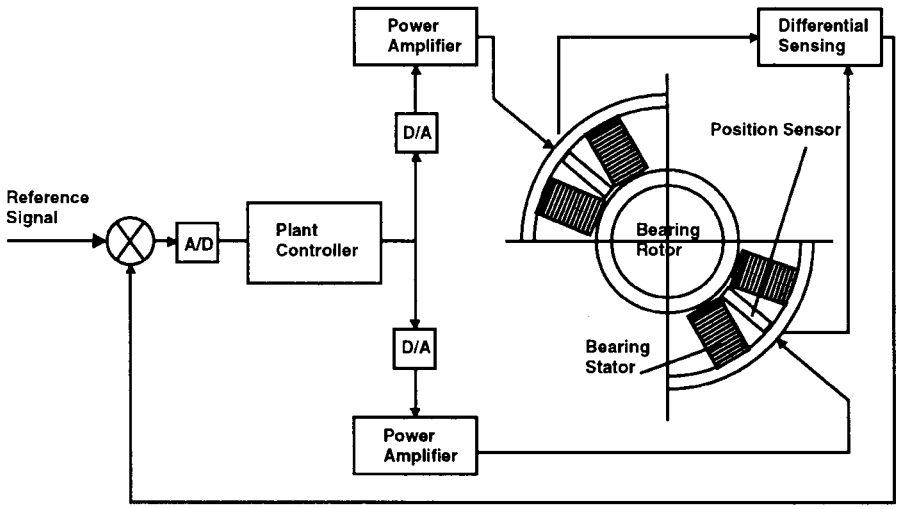


FIG. B-6 Schematic for a magnetic bearing. (Source: Revolve.)

The control process begins by measurement of rotor position with a *position sensor*. The signal from this device is received by the control electronics, which compare it to the desired position programmed during machine testing. Any difference between these two signals results in calculation of the forces necessary to pull the rotor back into the desired position. This is translated into a command to the *power amplifier* connected to the magnet actuator. The current is increased, causing an increase in magnetic flux, an increase in the forces between the rotating and stationary components, and, finally, movement of the rotor toward the stator along the *axis of control*.

The entire process is repeated thousands of times every second, enabling precise control of machinery rotating at speeds in excess of 100,000 rpm.

Actuators and sensors. This represents closed-loop control of one axis on a radial bearing. To provide support in more than one direction, magnetic poles are oriented about the periphery of a radial bearing. This is shown in Fig. B-7A, along with a photograph of a radial magnetic bearing rotor and stator (Fig. B-7B).

Radial actuator construction is very similar to that of an electric motor, involving the use of stacked laminations of steel, around which power coils are wound. Stacked laminations are also used in the rotor to minimize eddy current losses, which cause heating and otherwise represent inefficiencies between power supplied to the bearings, and the useful work performed by the bearing system.

The sensors are also oriented about the periphery of the stator, usually inside a ring that is mounted adjacent to the actuator poles. The photograph shows the sensor ring mounted on the end face of the radial stator. Inductive sensors are used that measure the inductance of the gap between the sensor and the rotor laminations. Two measurements are taken for each radial axis and the rotor center position calculated by means of a bridge circuit.

A typical rotating machine will experience forces in both the radial and axial directions. Typically, a five-axis orientation of bearings is used, incorporating two radial bearings of two axes each and one thrust bearing. The orientation of axes is shown in Fig. B-8.

Thrust bearings provide a magnetic flux path in the axial direction, between two stators oriented on either side of a *thrust rotor*, or *disc*, mounted on the

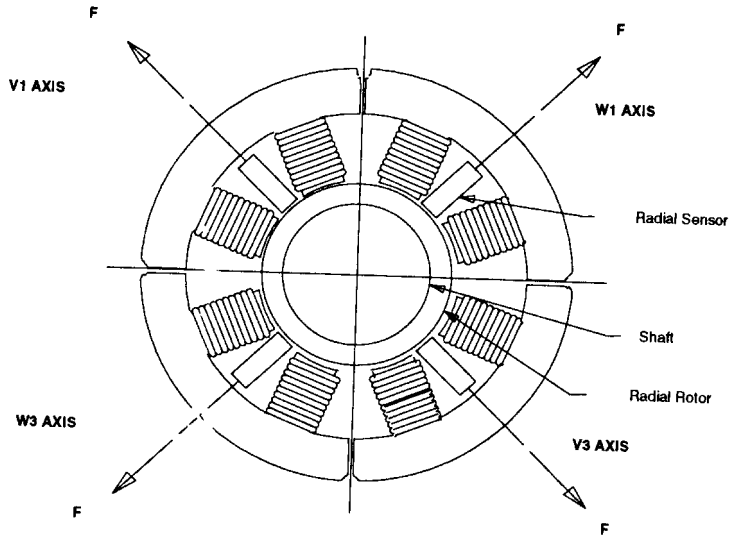


FIG. B-7A Magnetic poles oriented about the periphery of a radial bearing. (Source: Revolve.)

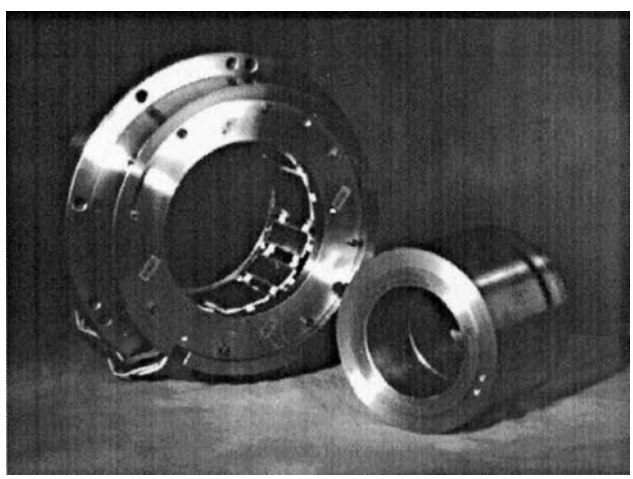


FIG. B-7B Radial magnetic bearing rotor and stator. (Source: Revolve.)

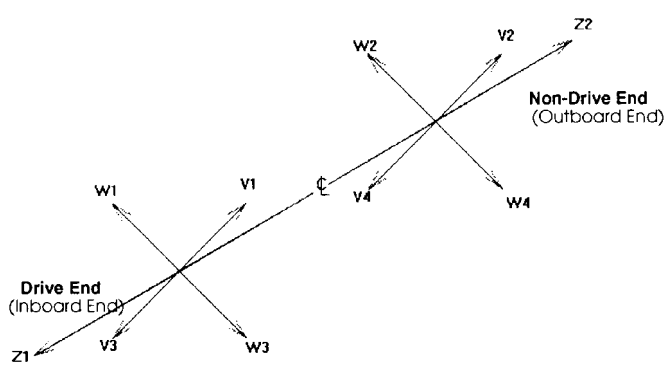


FIG. B-8 Vector diagram of bearing forces. (Source: Revolve.)

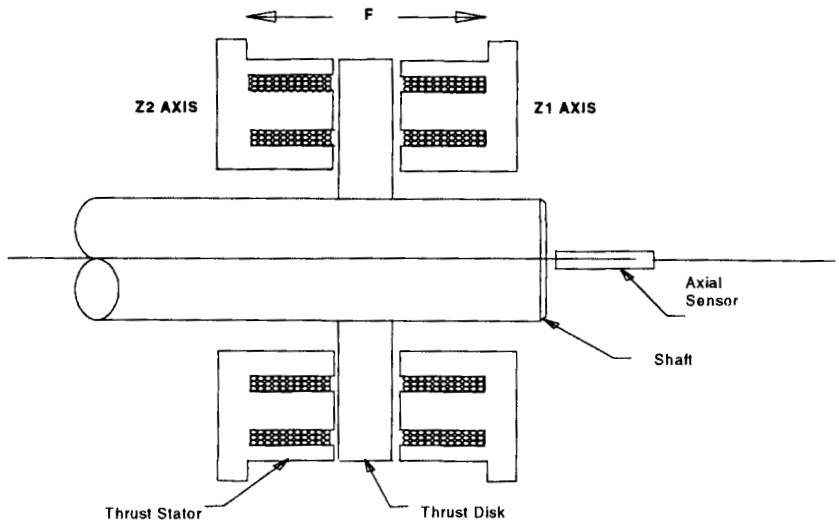


FIG. B-9A Thrust bearing schematic. (Source: Revolve.)

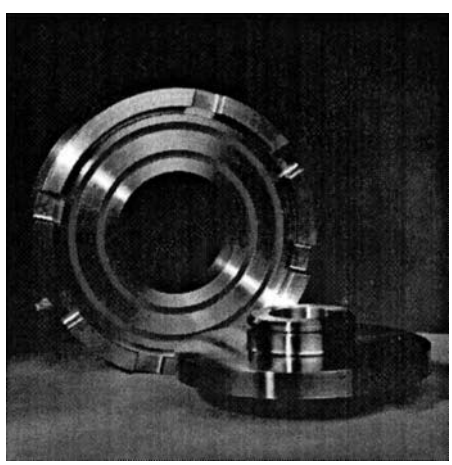


FIG. B-9B Thrust bearing. (Source: Revolve.)

rotating shaft as shown in Fig. B-9. An axial sensor measures the position of the shaft.

Control system. The control system consists of the *plant controller* (or control head), *power amplifiers*, and a *power supply*. Additional circuitry is present for conditioning of the signals from the position sensors and conversion of the digital



FIG. B-10A Low load digital control system. (Source: Revolve.)



FIG. B-10B High load digital control system. (Source: Revolve.)

outputs from the plant controller to the analog amplifiers. Finally, a *user interface* allows input of the desired rotor position information and logic to be coordinated with other machine systems (startup, warmup, shutdown, etc.).

Control systems for magnetic bearing systems can be of the analog or digital type. *Analog control systems* have been used in magnetic bearing control for over 30 years, but are rapidly being displaced by digital control systems, suitable for low load applications (Fig. B-10A) and high load (Fig. B-10B).

Low load control systems provide as much as 1 kW of power for applications such as turbomolecular pumps in an enclosure about the size of a shoe box. Higher load

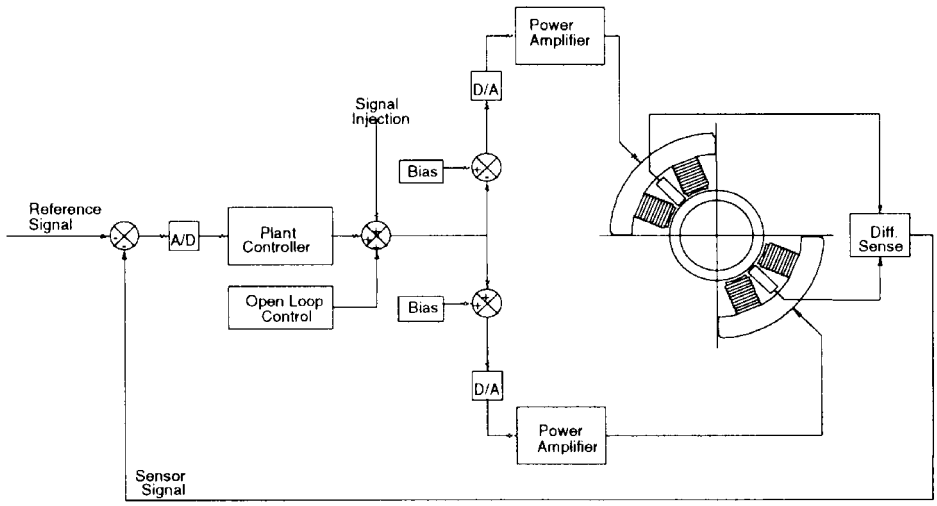


FIG. B-11 Open-loop control system. (Source: Revolve.)

applications include large-frame turbomachinery, such as centrifugal compressors, pumps, and turbines, and require up to 10 kW of power. These systems are designed to meet industrial electronic rack standards.

Control algorithm. For any particular bearing system, a control scheme is required that takes into account the behavior of the suspended (rotating) weight over the operating range (speed, acceleration, dynamic forces). These characteristics are revealed during a *rotor dynamic analysis* of the system. The knowledge gained results in composition of a *transfer function* for the control system that enables stability to be achieved over the desired operating range.

This transfer function is present in analog control systems in the form of hard-wired control circuits, which must be designed and built for each specific purpose or application. Their tolerance to system changes is limited, and modifications or changes are difficult and expensive to make.

Digital control systems allow transfer functions to be programmed in software, which can be modified or changed more easily and for less expense. They also allow *open-loop control* schemes to be used; these are control schemes that use actively changing external signals, as well as the reference and sensor signals used in closed-loop systems. (See Fig. B-11.)

Advantages and benefits

1. *Noncontacting*. Separation of the rotating and stationary parts means that the only friction is that provided by the air or other gas present. This is negligible compared to conventional oil or grease lubricants. Machinery operates more efficiently, with virtually no wear, requiring less maintenance and therefore achieving higher reliability.
2. *Contamination free*. Magnetic bearings can be used in processes that will not withstand the presence of any lubricant—such as refrigeration compression, pure gas processing, and vacuum processes such as those used in semiconductor manufacturing.
3. *Higher rotational speeds*. Noncontacting also means that higher rotational speeds are possible, as friction and surface galling are no longer factors to be

concerned with. Speeds are limited only by material stresses. This has significant implications for applications such as machine tool spindles, with potential for higher productivity and better surface finish quality.

4. *Power consumption*. Power consumed by a magnetic bearing system is minimal. Even for large turbomachinery, involving radial bearings supporting rotating assemblies of up to 15,000 N, power consumption is less than 4 kW. This is because the power is transferred back and forth between the bearing coils, and capacitors that straddle the power amplifiers. The only real power consumed is in resistance losses of the bearing coils and control system conductors, and eddy current losses in the bearing laminations. Compare this to over 250 kW of power consumed by a conventional system, which represents the sum of the lube oil shear losses, pump, and cooling fan power.

Blades and Vanes (for a Turbine) (*see* Metallurgy; Turbines)

Boilers

In conventional plants today, the term *boiler* is used synonymously with steam generator, which covers the boiler, the fuel system, and fuel firing mechanisms and controls, as well as all other accessories and controls.

Boiler Emissions

For allowable emissions limits in a sample country, *see* Emissions.

Reference and Additional Reading

1. Soares, C. M., *Environmental Technology and Economics: Sustainable Development in Industry*, Butterworth-Heinemann, 1999.

Borescopes

A *borescope* is an instrument that, basically, uses fiber optics and flexible hose to "see" inside a critical piece of machinery. The objective with machinery is generally to run it "on condition" for as long as possible or, in other words, to use predictive instead of preventive maintenance (*see* Condition Monitoring). The borescope helps one do this by allowing maintenance personnel to look at the inside of a critical item of machinery. The OEM designers normally provide appropriate ports for this look. A qualified observer can then determine items such as: is there evidence of overtemperature on the turbine blades, are combustor can cross-rods cracking, is there erosion on fuel nozzles, and so forth? There is some distortion afforded by the fish eye at the end of the borescope's flexhose, but seasoned observers can interpret what they see. Interpreting borescope views is more an art than a detailed science.

Brakes

Hydrodynamic*

Hydrodynamic brakes are similar in construction to hydrodynamic couplings. The force or torque delivered by the machine to be braked is applied to the pump

*Source: J.M. Voith GmbH, Germany. Adapted with permission.

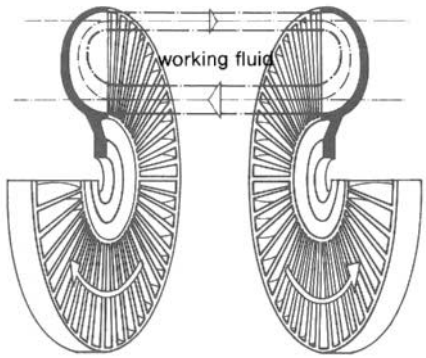


FIG. B-12 Hydrodynamic brake schematic. (Source: J.M. Voith GmbH.)

wheel and from there, by way of the working fluid, to the turbine wheel. (See Fig. B-12.)

The difference in the case of hydrodynamic brakes is that the turbine wheel is not free to rotate.

In order to achieve the highest possible braking force, the blades of the rotor and stator are set at an angle. The hydrodynamic brake model discussed in this section (made by Voith) achieves its braking power by retarding the inertial force of the working fluid that is accelerated radially and tangentially in the driven part (rotor) and decelerated in the nonrotating part (stator). The kinetic energy is thus transformed into heat that is taken up by the working fluid.

The braking process is initiated by filling the brake with working fluid and is ended by draining the fluid from the brake.

By varying the amount of fluid in the brake or the filling pressure, infinite variation of the braking torque within the performance range of the brake concerned can be obtained.

It is possible to influence the shape of the performance curve by various control and regulation means.

Control and energy dissipation

In hydrodynamic brakes, the working fluid is used simultaneously as a braking medium and as an energy carrier. In order to dissipate the heat energy generated by the braking process, a suitably dimensioned flow of working fluid with a defined input temperature must be ensured at all times.

A hydrodynamic brake can either be controlled with the working fluid in closed circuit or in open circuit and, by adjusting the amount of fluid in the brake, it is possible to work at any point within the performance range of the brake concerned.

A closed-circuit hydrodynamic brake (Fig. B-13) is directly connected with one or more heat exchangers and a given amount of working fluid once introduced into the system circulates continuously between the brake and the heat exchanger. To change the amount of fluid in the brake, fluid is either added to or withdrawn from the circuit by way of valves. This alters the pressure within the system and establishes the required performance within the working range.

With an open-circuit hydrodynamic brake (Fig. B-14) the working fluid is fed to the brake from a tank by means of a pump. The required amount of fluid in the brake is maintained by a control element in the feed line.

The heated working fluid flows back to the tank through the return line and a second circuit provides for cooling the fluid in the storage tank.

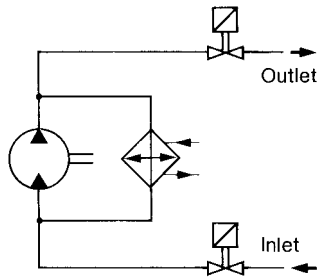


FIG. B-13 Closed-circuit hydrodynamic brake. (Source: J.M. Voith GmbH.)

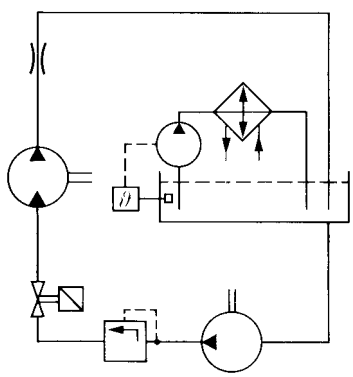


FIG. B-14 Open-circuit hydrodynamic brake. (Source: J.M. Voith GmbH.)

Where water is used as a working fluid it is possible for the water to be drawn direct from the water mains and for it to be discharged to the drains after use.

Hydrodynamic brakes are used wherever substantial masses have to be braked without the braking mechanism being subject to wear and where mechanical brakes would be overtaxed. Hydrodynamic brakes are particularly well suited to applications involving continuous operation. Hydrodynamic brakes have been used over several decades as a means of measuring power, particularly as absorption dynamometers (fluid friction brakes) in engine test stands.

Applications for hydrodynamic brakes

Test stands of all types, for instance

- Axle test stands
- Torque converter test stands
- Engine test stands
- Transmission test stands
- For the testing of hydrostatic units
- Ski slope servicing vehicle test stands

Coal crushers

Protective brakes in steel hydraulic mechanisms

Gas turbine drives

Deep drilling rigs
Downward sloping belt conveyors
Winding engine construction
Hydrostatic drives
Shipbuilding
Braked trailers (load simulators)

Advantages of hydrodynamic brakes

Simple construction
No fading
Virtually maintenance-free
Long service life
Low weight for high capacity
Small rotating masses and small inertia moment
Wear-free operation
Highly economic
Infinitely variable braking torque
Simple control options
High degree of stability of the set working point
Low and controllable working temperatures

Specific hydrodynamic brake models (VHBI, VHBM*: construction and variants)

VHBI hydrodynamic brakes. These brakes are characterized by performance with small overall dimensions as a result of the inclined blading of the rotor and stator. The hydrodynamic brake is effective in only one direction of rotation.

The housing elements, rotor, and stator are of spheroidal graphite iron or of material that is resistant to cavitation and corrosion. The shaft is of high-quality, heat-treatable steel and the shaft extensions are normally finished to DIN standards although other arrangements are possible. Oil is used as a lubricant for the bearings and radial-lip-type shaft seals or labyrinth seals are normally fitted. Mechanical end-face seals can be provided.

VHBM hydraulic brakes for power measurement. The VHBM is similar in construction and execution to the VHBI. The braking torque is transmitted through the housing, which is free to move by way of a lever to an electrical, pneumatic, or mechanical force measuring device while the speed of the shaft is detected at the same time and peripheral units provide for registering, processing, indicating, and recording the measurement data. Suitable control devices provide for adapting the brake characteristics to the application concerned.

Non-self-supporting models. Where the hydrodynamic brake is fitted in a machine or drive element, use is made of the existing bearings. The rotor is fitted overhung on the shaft and the braking torque is transmitted to the machine housing.

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*VHBI and VHBM are trademark model designations associated with Voith brakes.

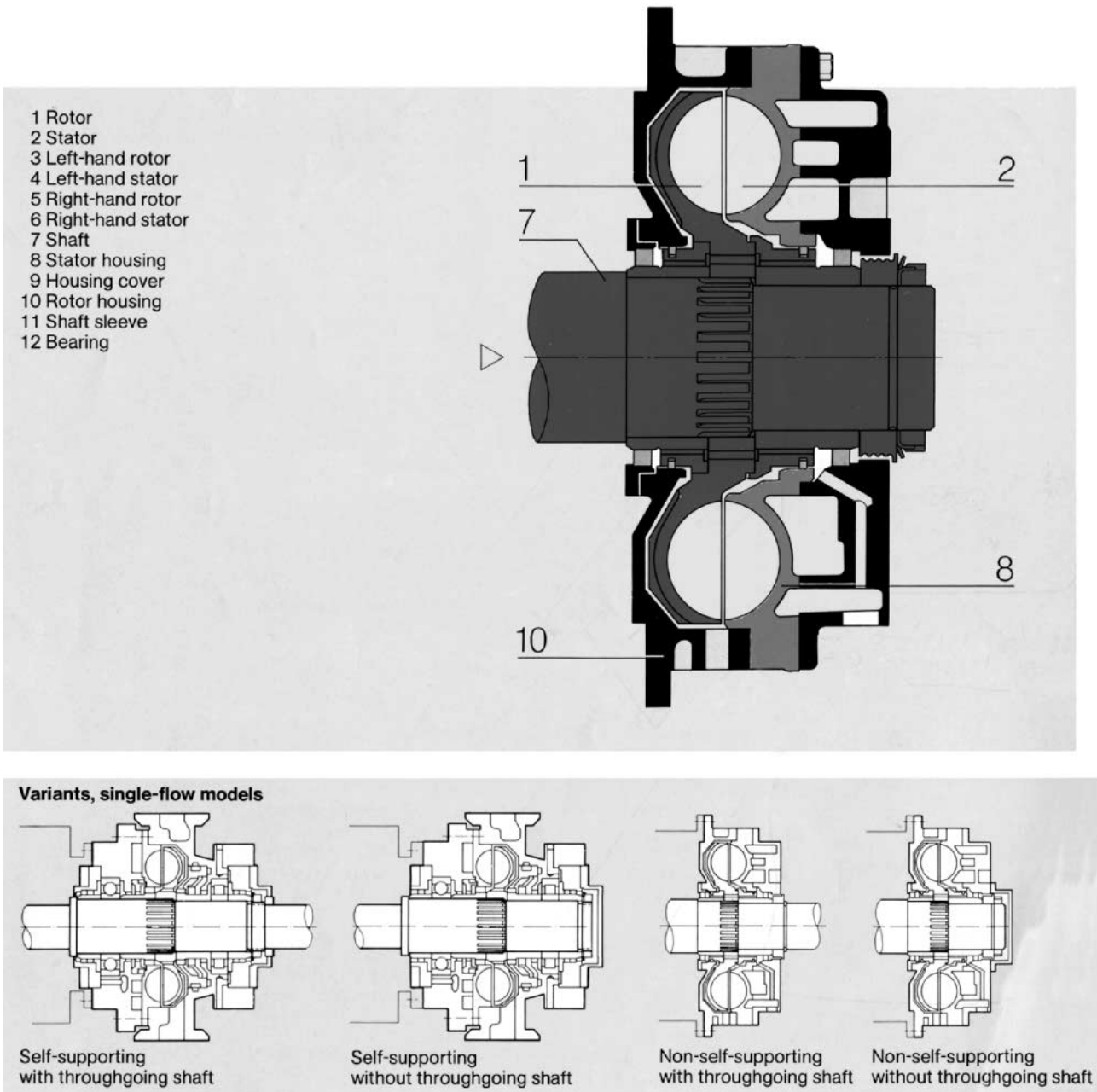


FIG. B-15 Hydrodynamic brake components and brake types (Voith VHBM type). (Source: J.M. Voith GmbH.)

Self-supporting models. Here the hydrodynamic brake is provided with its own bearings—roller bearings or, if requested, plain bearings. The hydrodynamic brake is either flange-mounted or separately mounted on the base plate as one of the elements in a drive assembly. See Fig. B-15.

After reaching a certain profile diameter, hydrodynamic brakes are supplied as double-flow models. Such models are indicated by a letter "D" in the type designation.

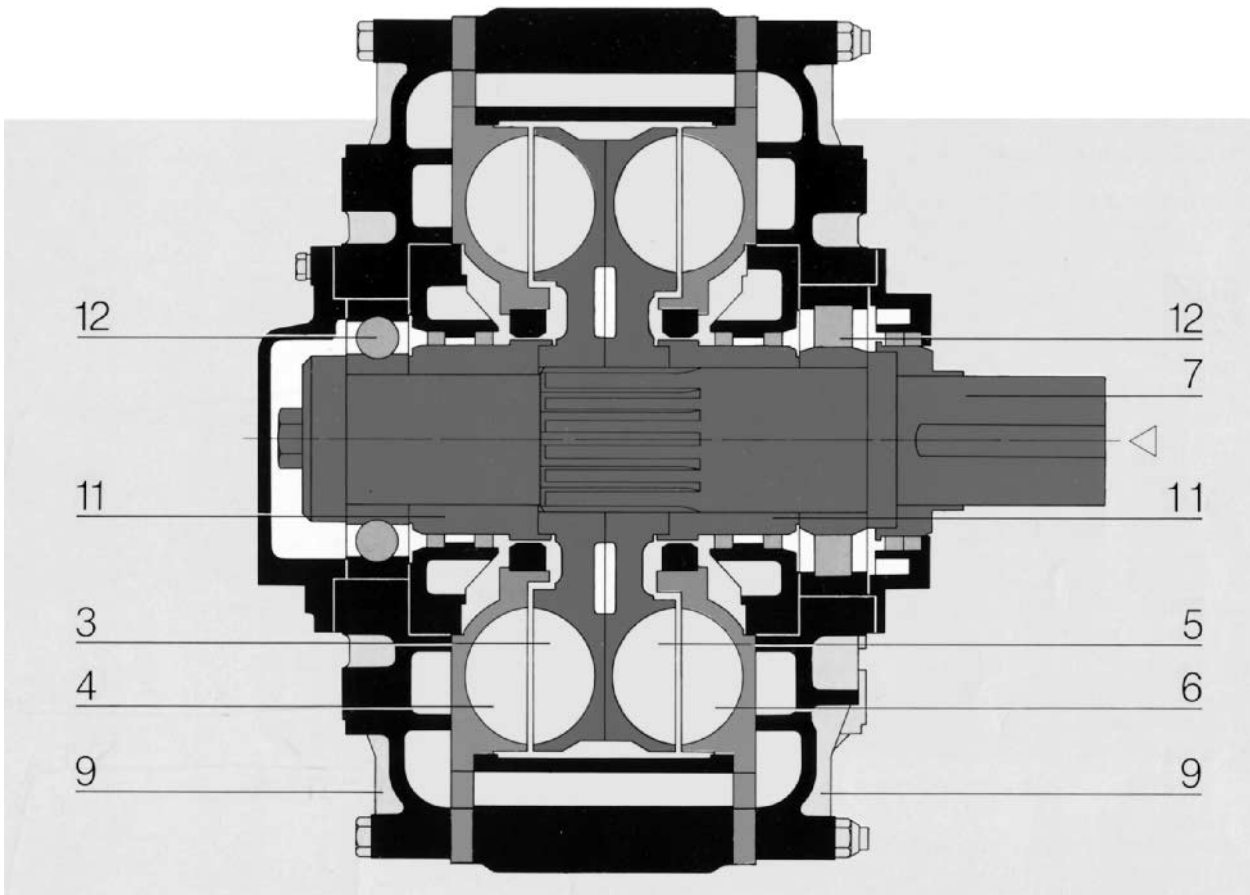


FIG. B-15 (*Continued*)

Controls for hydrodynamic brake types

The control schematics shown in Figs. B-16 through B-21 are some of the many practical applications of controls for hydrodynamic brakes.

Performance—braking torque

Separate performance characteristics are indicated in Figs. B-22 through B-25 for brakes using water and oil as a working medium.

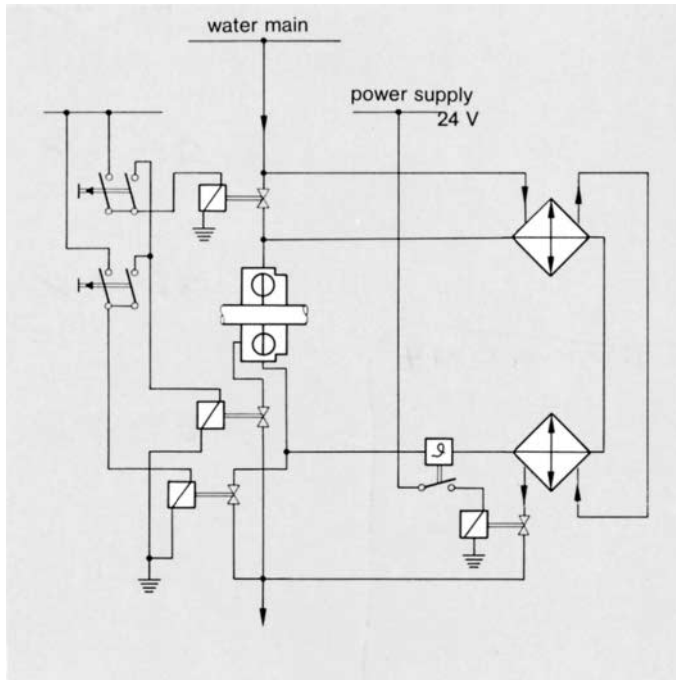


FIG. B-16 Heat-exchanger control. The supply is from the water main. Solenoid valves control the extent to which the closed circuit is filled. In order to reduce the water consumption, the heat energy is dissipated through a heat exchanger through which the flow is controlled in relation to the temperature. (Source: J.M. Voith GmbH.)

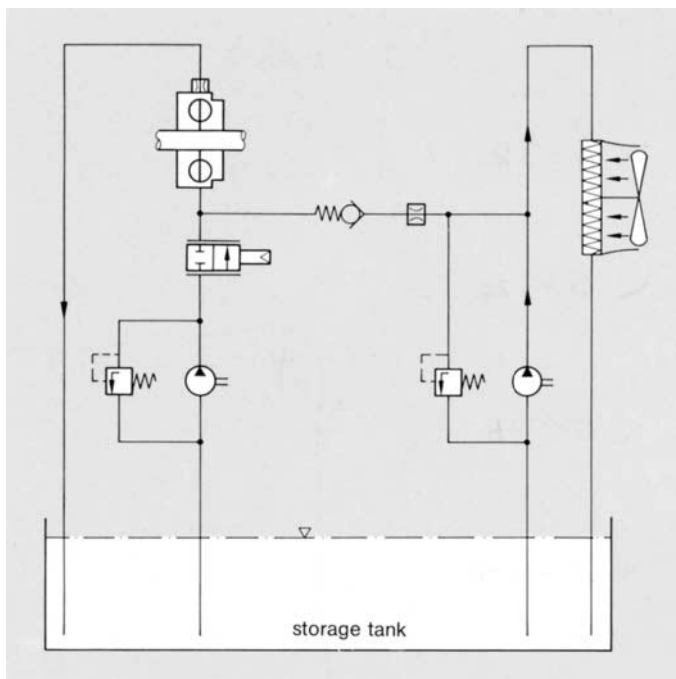


FIG. B-17 Pump control with separate cooling for the storage tank. The working fluid is drawn from the storage tank and fed to the hydrodynamic brake by means of a pump and by way of a pressure relief valve—to maintain the pressure at a constant level—and a solenoid valve—to turn the supply on and off. The heated working fluid flows back to the storage tank and is cooled in a separate circuit that also provides for the constant cooling and lubrication of the hydrodynamic brake. (Source: J.M. Voith GmbH.)

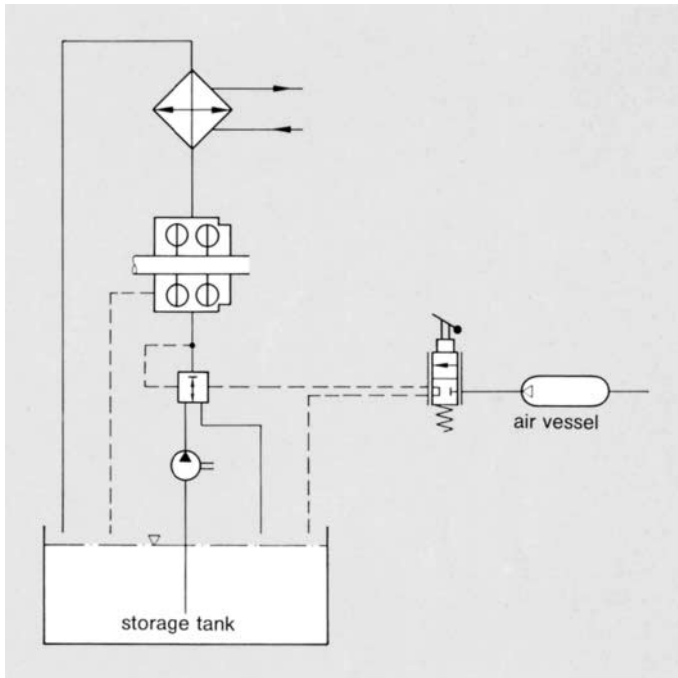


FIG. B-18 Pump control with integral cooling. The supply to the hydrodynamic brake is effected by a pump by way of a pneumatically controlled pressure-regulating valve. The heated working fluid is cooled in a heat exchanger in the return line. (Source: J.M. Voith GmbH.)

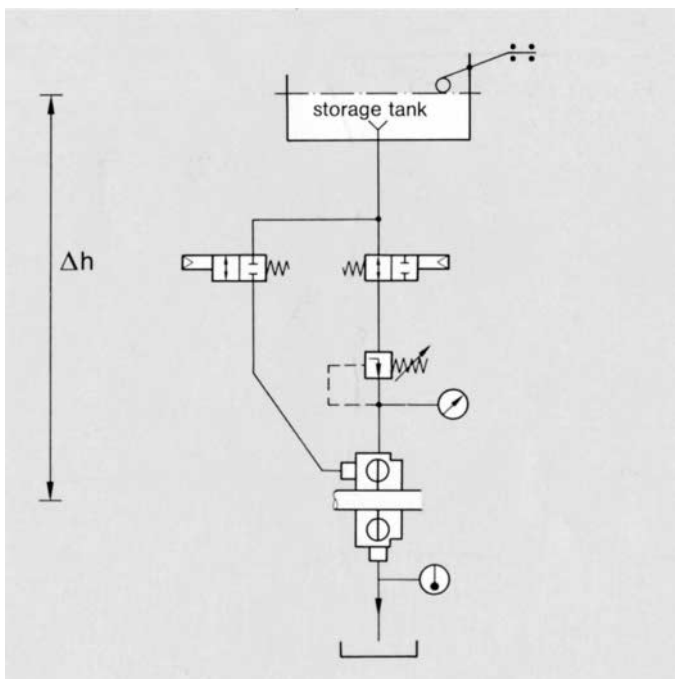


FIG. B-19 Through-flow control. Here the hydrodynamic brake is fed with water from a water main or from an elevated storage tank by way of a pressure-regulating valve and solenoid valve. The heated water is discharged to a drain. (Source: J.M. Voith GmbH.)

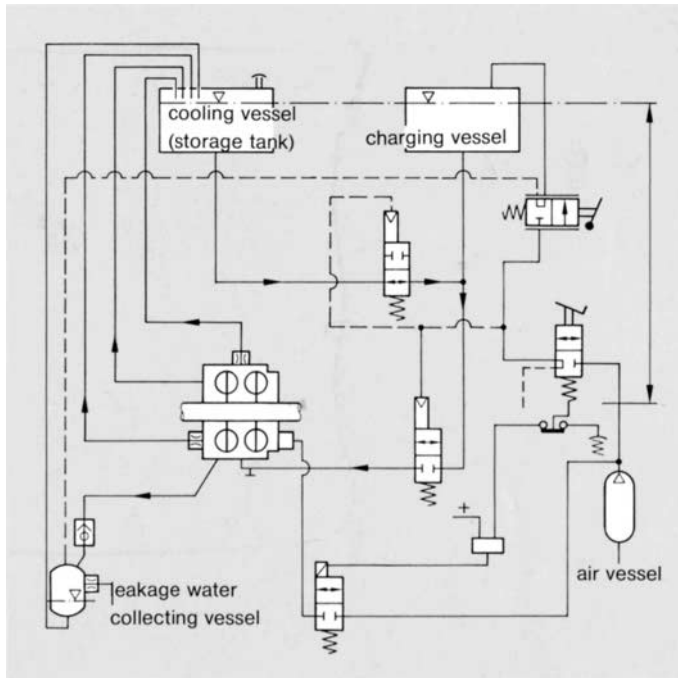


FIG. B-20 Air vessel control—with capacitative heat storage in the storage tank. The charging vessel is filled by gravity. On braking, the connection between the two vessels is interrupted. Compressed air is applied to the charging vessel and the working fluid is forced into the brake. The filling pressure is manually preselected. The heated working fluid flows back into the cooling vessel. Any leakage water flows into a separate collecting vessel and is then fed back to the cooling vessel by applying compressed air. (Source: J.M. Voith GmbH.)

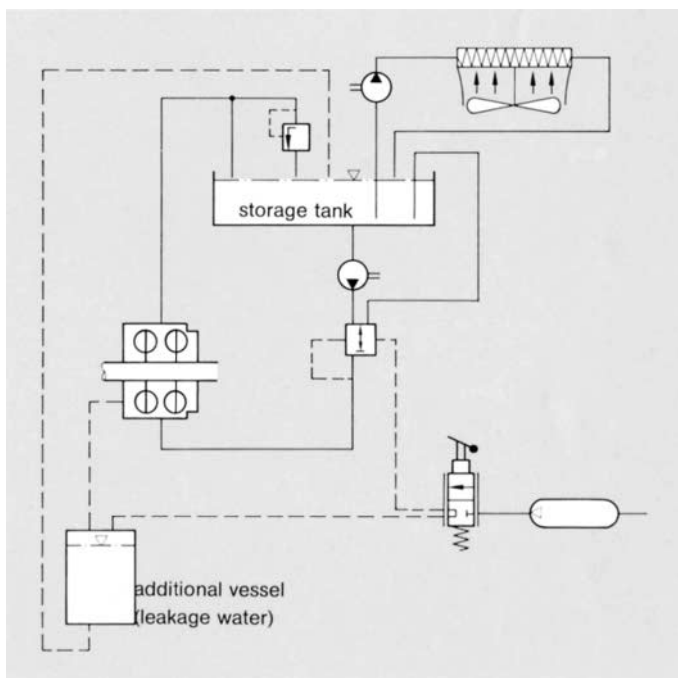


FIG. B-21 Pump control with separate cooling circuit. The hydrodynamic brake is supplied with fluid by a pump by way of a pneumatically controlled pressure-regulating valve. An additional vessel is provided to collect any leakage water, which is then forced back into the storage vessel by compressed air. The storage vessel is cooled by way of a separate cooling circuit. (Source: J.M. Voith GmbH.)

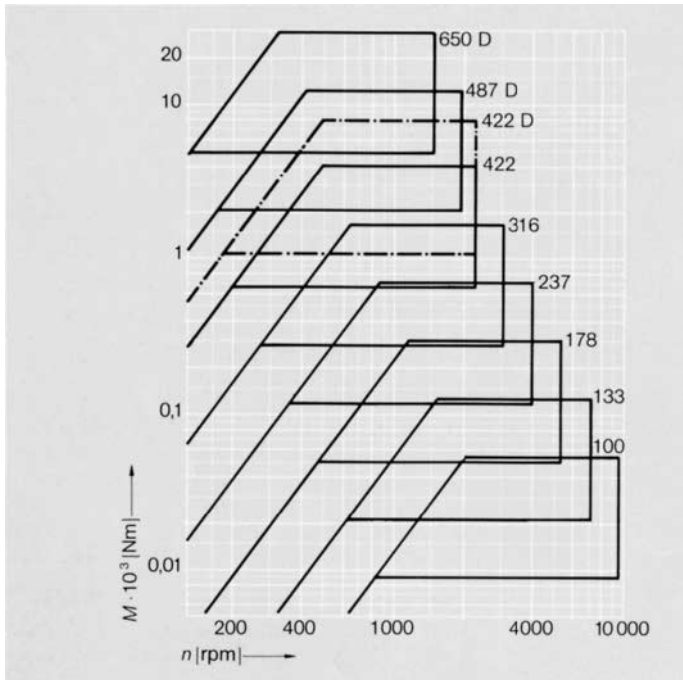


FIG. B-22 $M = f(n)$, **braking torque as a function of speed. (Source: J.M. Voith GmbH.)**

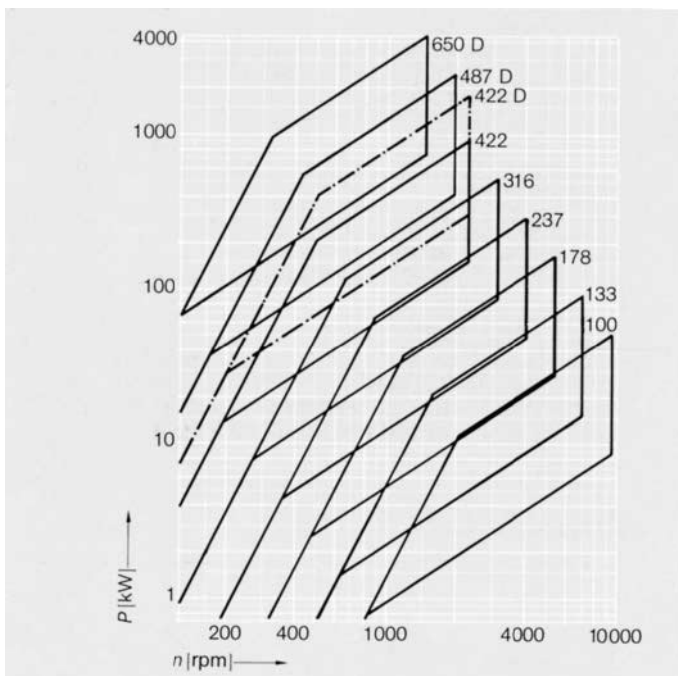


FIG. B-23 $P = f(n)$, **braking power as a function of speed. (Source: J.M. Voith GmbH.)**

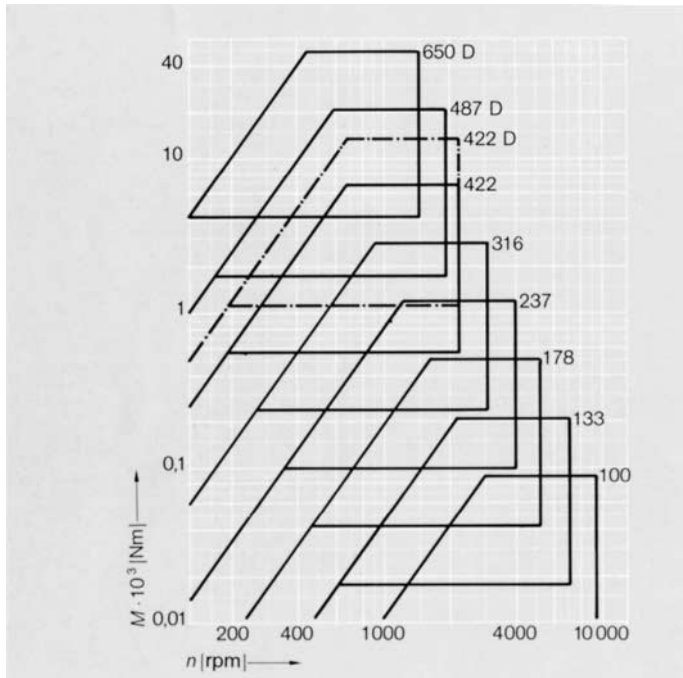


FIG. B-24 $M = f(n)$, **braking torque as a function of speed. (Source: J.M. Voith GmbH.)**

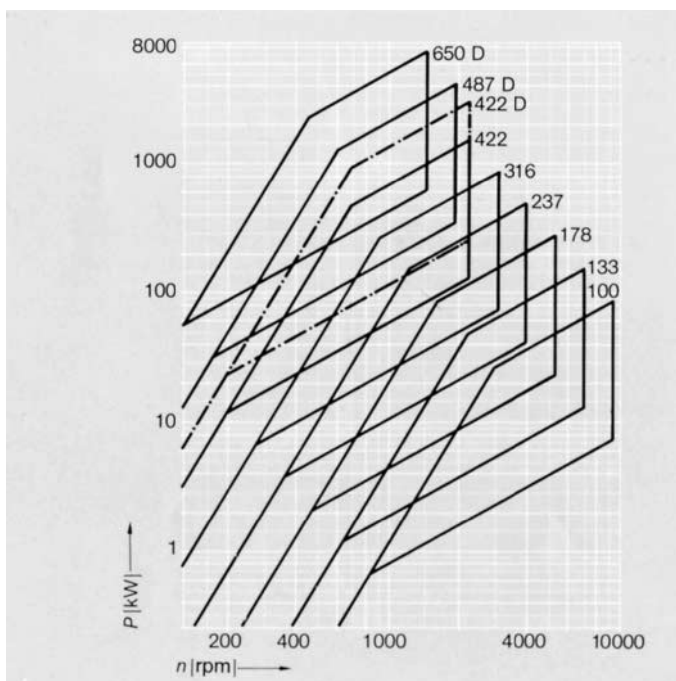
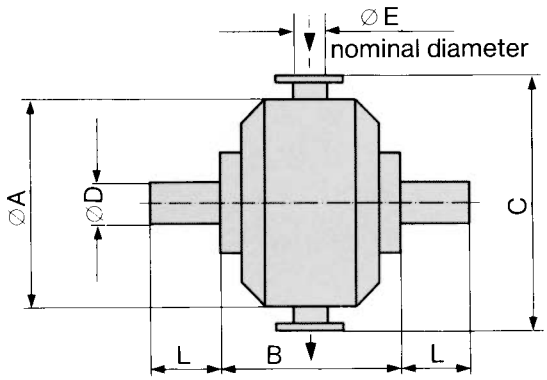


FIG. B-25 $P = f(n)$, **braking power as a function of speed. (Source: J.M. Voith GmbH.)**

Single-flow models



Double-flow models

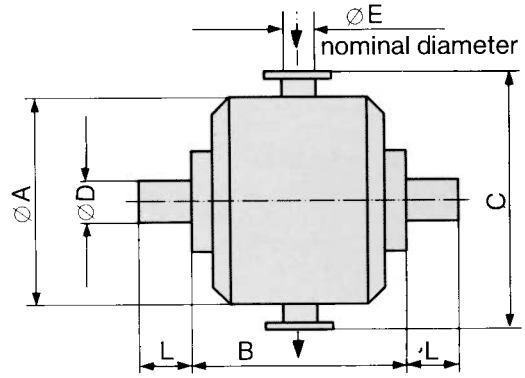


FIG. B-26 Reference dimensions of different brake models. (Source: J.M. Voith GmbH.)

Dimensions. See Fig. B-26 for a diagrammatic definition of reference dimensions.

Single-Flow Models

Brake Type	A (mm)	B	C	D	E	L	Weight (kg)
100	130	130	150	28	20	42	6
133	170	170	195	30	25	58	13
178	225	225	260	40	32	82	32
237	300	300	345	55	40	82	76
316	400	400	460	75	65	105	180
422	535	535	615	95	80	130	430

All dimensions and weights given represent maximum values. The dimensions "D" and "L" represent the dimensions necessary for the transmission of the indicated maximum torque.

Double-Flow Models

Brake Type	A (mm)	B	C	D	E	L	Weight (kg)
100 D	125	133	150	30	25	58	7
133 D	170	176	195	38	32	58	15
178 D	225	236	260	55	50	82	37
237 D	300	315	345	70	65	105	88
316 D	400	420	460	90	80	130	208
422 D	535	560	615	120	100	165	495
487 D	620	646	710	140	125	200	761
650 D	825	865	950	190	150	280	1809
826 D	1050	1100	1210	240	200	330	3712

All dimensions and weights given represent maximum values. The dimensions "D" and "L" represent the dimensions necessary for the transmission of the indicated maximum torque.

Technical data

Single-Flow Models

Brake Type	M_{BR} max. (N·m)	n Abreg. (min^{-1})	n max. (min^{-1})
Working medium: oil			
100	86	2,868	9,549
133	203	2,164	7,180
178	484	1,620	5,365
237	1,157	1,215	4,029
316	2,712	910	3,022
422	6,494	684	2,263
Working medium: water			
100	52	2,037	9,549
133	121	1,541	7,180
178	291	1,151	5,365
237	681	869	4,029
316	1,634	652	3,022
422	3,831	490	2,263

Double-Flow Models

Brake Type	M_{BR} max. (N·m)	n Abreg. (min^{-1})	n max. (min^{-1})
Working medium: oil			
100 D	172	2,868	9,549
133 D	406	2,164	7,180
178 D	968	1,620	5,365
237 D	2,314	1,215	4,029
316 D	5,424	910	3,022
422 D	13,032	684	2,263
487 D	20,000	598	1,961
650 D	47,863	444	1,469
826 D	97,067	349	1,156
Working medium: water			
100 D	104	2,037	9,549
133 D	242	1,541	7,180
178 D	582	1,151	5,365
237 D	1,362	869	4,029
316 D	3,268	652	3,022
422 D	7,662	490	2,263
487 D	12,069	423	1,961
650 D	28,292	316	1,469
826 D	64,050	249	1,156

Control characteristic curves—examples of application

Controls can be provided to give various characteristic curve forms. The characteristic curve form should be so selected that load moment line intersects the brake torque line at a defined point. (See Figs. B-27 and B-28.) Figure B-29 shows a brake assembly schematic. Figures B-30 through B-36 show applications of brake assemblies in various industries.

Brick-Lined Process Equipment

The thermal insulation value and other properties of the brick and protective membrane layer between the brick and the process fluid it contains can be determined from property values (such as corrosion resistivity) listed in catalogs supplied by the brick manufacturers or relevant engineering standards. *See* Some Commonly Used Specifications, Codes, Standards, and Texts.

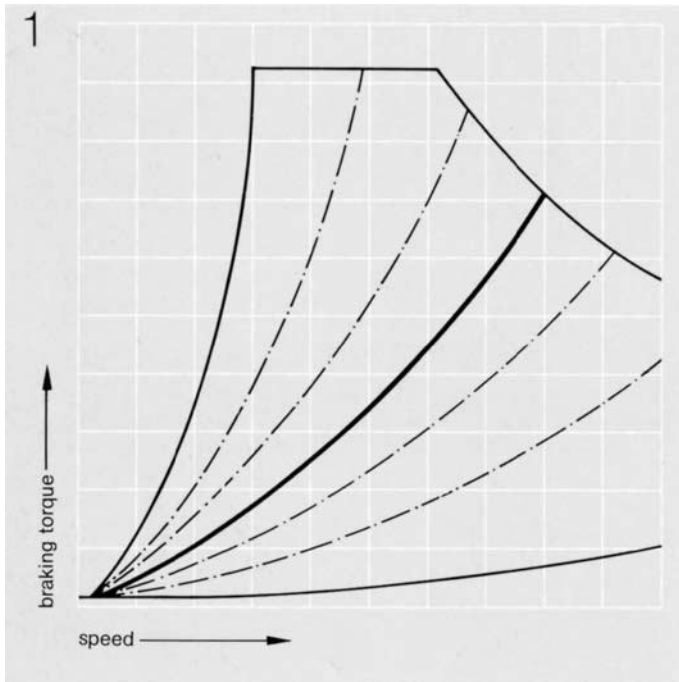


FIG. B-27 Control for a parabolic braking torque. (Source: J.M. Voith GmbH.)

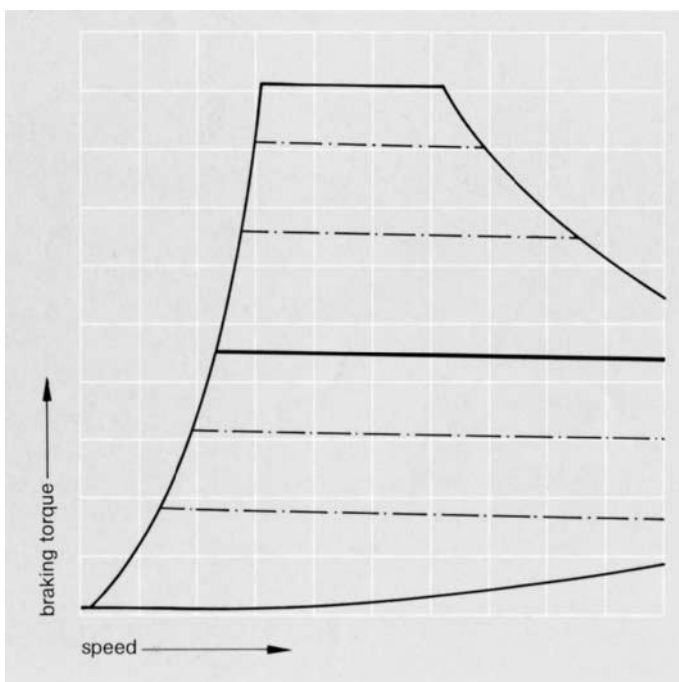


FIG. B-28 Control for a constant braking torque. (Source: J.M. Voith GmbH.)

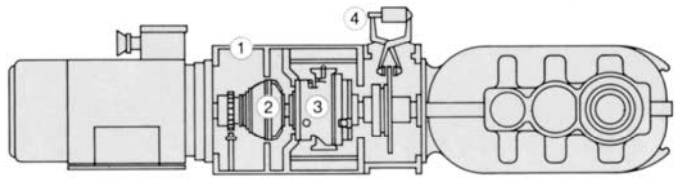


FIG. B-29 Schematic of a brake assembly. 1, connecting housing; 2, turbo coupling TV; 3, hydrodynamic brake VHBI; 4, mechanical brake. (Source: J.M. Voith GmbH.)



FIG. B-30 VHBI 422 D, fitted in the drive to the winch of a transportable drilling rig suitable for drilling to a depth of approximately 5000 meters, Texas, USA. (Source: J.M. Voith GmbH.)

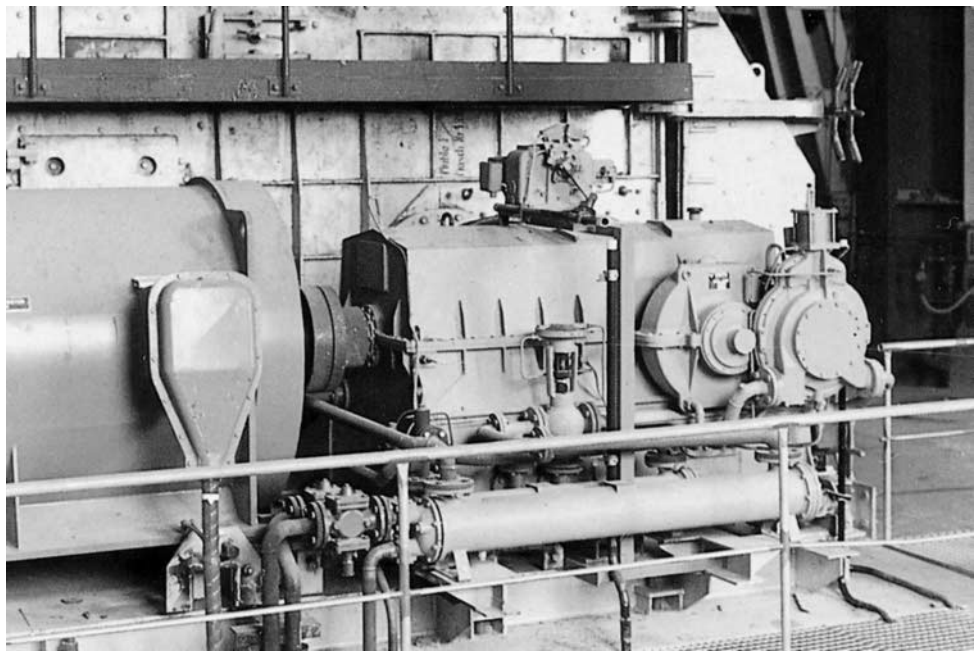


FIG. B-31 VHBI 316 with stepup gear, flange-mounted to a Voith variable-speed geared turbo coupling R 750 B in the drive to a coal crusher in a power station in West Germany. (Source: J.M. Voith GmbH.)

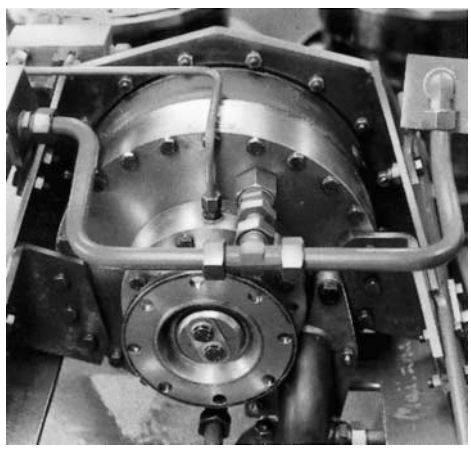


FIG. B-32 VHBI 316 working in connection with a Voith coupling 562 SVL as an element in a performance test stand for electric motors in Italy. (Source: J.M. Voith GmbH.)

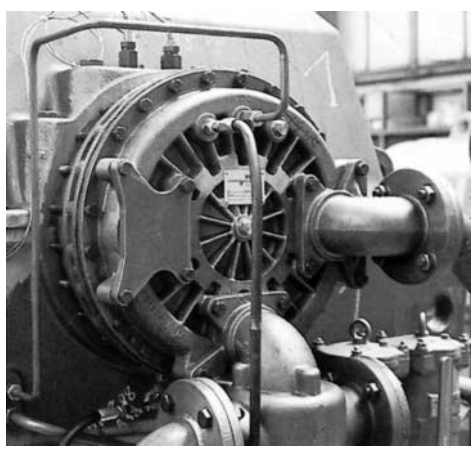


FIG. B-33 VHBI 422 flange-mounted to a Voith variable-speed geared turbo coupling R 866 B 5 in the drive to a coal crusher in a power station in Yugoslavia. (Source: J.M. Voith GmbH.)

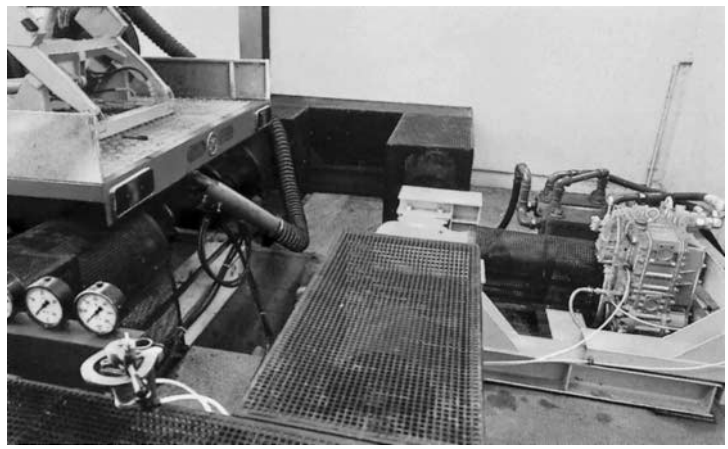


FIG. B-34 VHBK 130 fitted in a test stand for testing the hydrostatic drive system of ski slope maintenance machines in West Germany. (Source: J.M. Voith GmbH.)

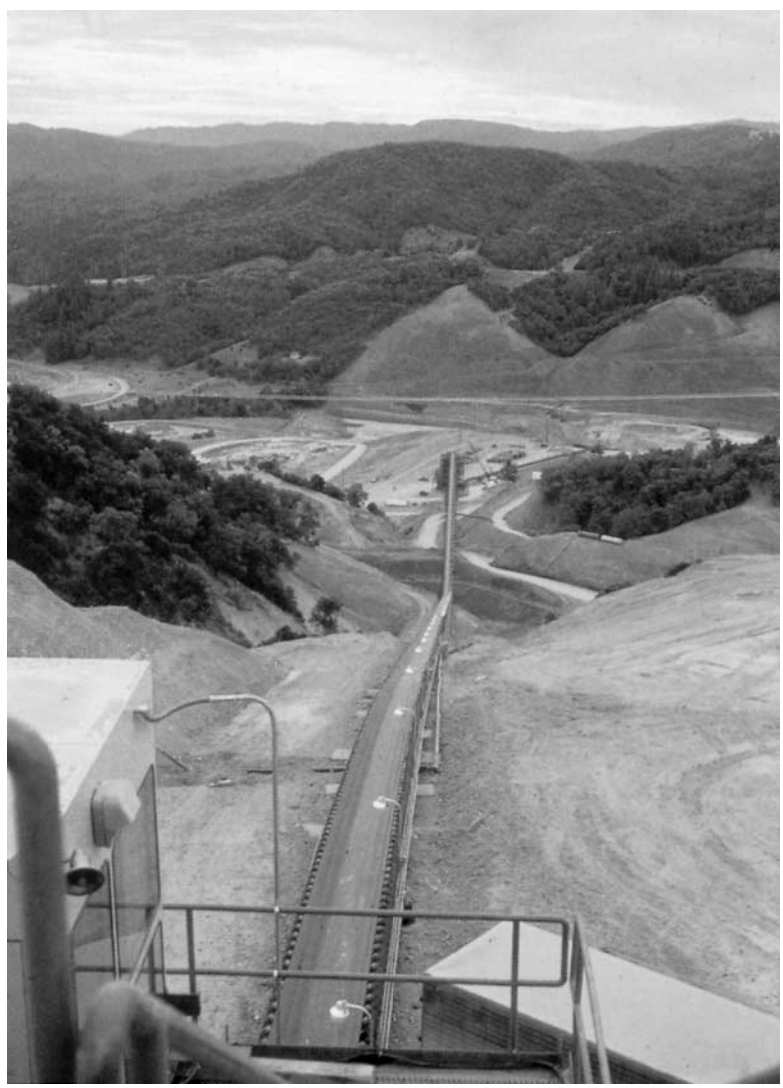


FIG. B-35 VHBI 422 with throughgoing shaft, fitted to a Voith variable-speed turbo coupling 650 SVNL in the drive to a 1-km-long downward sloping conveyor belt feeding material to an imponding dam in California, USA. (Source: J.M. Voith GmbH.)



FIG. B-36 VHBI 316 with stepup gear, flange-mounted to a reduction gear in one of the drive stations in a 10-km-long conveyor plant, made up of several conveyor belts, for the conveyance of bauxite in West Australia. (Source: J.M. Voith GmbH.)

Briquette Machines (Similar to Pelletizing and Balling Machines)

These types of machinery are custom built for different products and processes. Basically, work energy is used to make briquettes of varying size and shape. These machines are common in agricultural, domestic, and chemicals industries. *See* Some Commonly Used Specifications, Codes, Standards, and Texts.